

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and built **test components** for engineering evaluation, translating intangible design inputs into tangible engineering specifications and drawings for a medical device.
- Assisted in the development and optimization of **process development** activities, understanding key process inputs and outputs using **statistical methods** such as DOE.
- Prepared **work instructions** and **standard operating procedures (SOPs)**, and drafted technical documents and reports for R&D; and Quality Assurance teams.
- Developed new **engineering test methods** and simulated use test methods, preparing and assisting in the validation of these methods.
- Performed lab bench testing and conducted engineering evaluations, including **development unit testing** for device components.
- Mentored R&D; technicians through new product development activities, specifically on **new process development** and materials inventory control.

SKILLS

Medical Device Development – Design Inputs, Design Specifications, Process Development, Test Method Development, Verification and Validation, IQ, OQ, PQ, cGMP, GLP

Engineering Tools & Methodologies – Design of Experiments (DOE), Statistical Analysis, Root Cause Analysis, Six Sigma, Design for Six Sigma

Software & Languages – Python, Java, SQL, Spring Boot, R, RStudio, MATLAB (implied by engineering context)

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Pandas, NumPy

Development Practices – Git, Jenkins, Jira, Confluence, REST APIs

PROJECTS

Medical Device Component Design & Validation Framework

- Designed and prototyped novel components for a medical device using **CAD software** and rapid prototyping techniques, focusing on material compatibility and mechanical integrity.
- Developed a suite of **engineering test methods** for evaluating component performance under simulated use conditions, including stress and fatigue testing, and documented these in formal protocols.
- Assisted in the validation of these test methods, ensuring their suitability for R&D; evaluation and pre-clinical studies, contributing to regulatory documentation packages.

Process Optimization for Medical Device Manufacturing

- Applied **Design of Experiments (DOE)** to optimize a key manufacturing process step, identifying critical process inputs and their impact on output quality for a device sub-assembly.
- Documented optimized process parameters and created draft **Standard Operating Procedures (SOPs)** for manufacturing technicians, ensuring consistent execution and product quality.
- Analyzed process data using **R** and generated reports detailing findings and recommendations for yield improvement, contributing to a **15% reduction** in process variability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)